Attack

Alternate Data Streams or "Doctor Jekyll and Mr. Hyde" Move to NTFS

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Difficulty

Introduced in Windows NT, the NTFS file system brought about a new concept: multiple streams into a single file known as Alternate Data Streams – abbreviated as ADS in this article. Supporting multiple streams of data into a single file within the NTFS file system is known by very few specialists and may prove to be underestimated in time.

The main characteristic of ADS is *invisibility*. When you read the content of a file you access only the main stream of that file. Any other streams attached to a file are invisible to file-handling utilities like Windows Explorer or the *Dir* command on MS-DOS. Many specialists complain about this feature because it can be exploited by a virus and very few people try to use it for beneficial purposes.

Enough with Theory, Let's Do Some Practice

In this example we utilize MS-DOS and Visual Basic Scripts. For clarification, the ADS name includes original file name, a colon and the ADS name. For example: `C:\Test\FileName.txt:StreamName.txt` Where `FileName.txt` is the main stream and `StreamName.txt` is the ADS. For all future examples we will use a new folder, `C:\TestADS`, where we will save all files and scripts. If you want to follow along, I suggest you create the same folder structure. In this example, I will show how to create a stream using Notepad: *Click Start, and then click Run* In the Open dialog box, type the following command:

```
Notepad.exe C:\TestADS\
   Filename.txt:StreamName.txt
```

A message box (Figure 1) will ask you if you want to create a new file. Press the *Yes* button and voila, you have created your first ADS. Type some text in document and close it. Confirm that you want to save the changes. Now, using Windows Explorer, open the file `C:\TestADS\Filename.txt`. You will notice that the size of the file is 0 bytes and the document is blank. To edit the ADS you must open it using the Run command. By using *Open* command in Notepad window and typing the address: `C:\TestADS\Filename.txt:StreamName.txt` you will receive an error message. So instead, click *Start* and then *Run* from the *Windows Start Menu*.

What you will learn...

- How to create, delete or view ADS
- How can we use the ADS for our own benefit
- What are malicious uses of ADS

What you should know...

- Visual Basic Script language

Type *notepad.exe* `C:\TestADS\Filename.txt:StreamName.txt`, in the open box or select its item from the list and then click *OK* to open it. Type some text, save the document and close it. The file `C:\TestADS\Filename.txt` will still remain 0 bytes in size, but the modified date has changed.

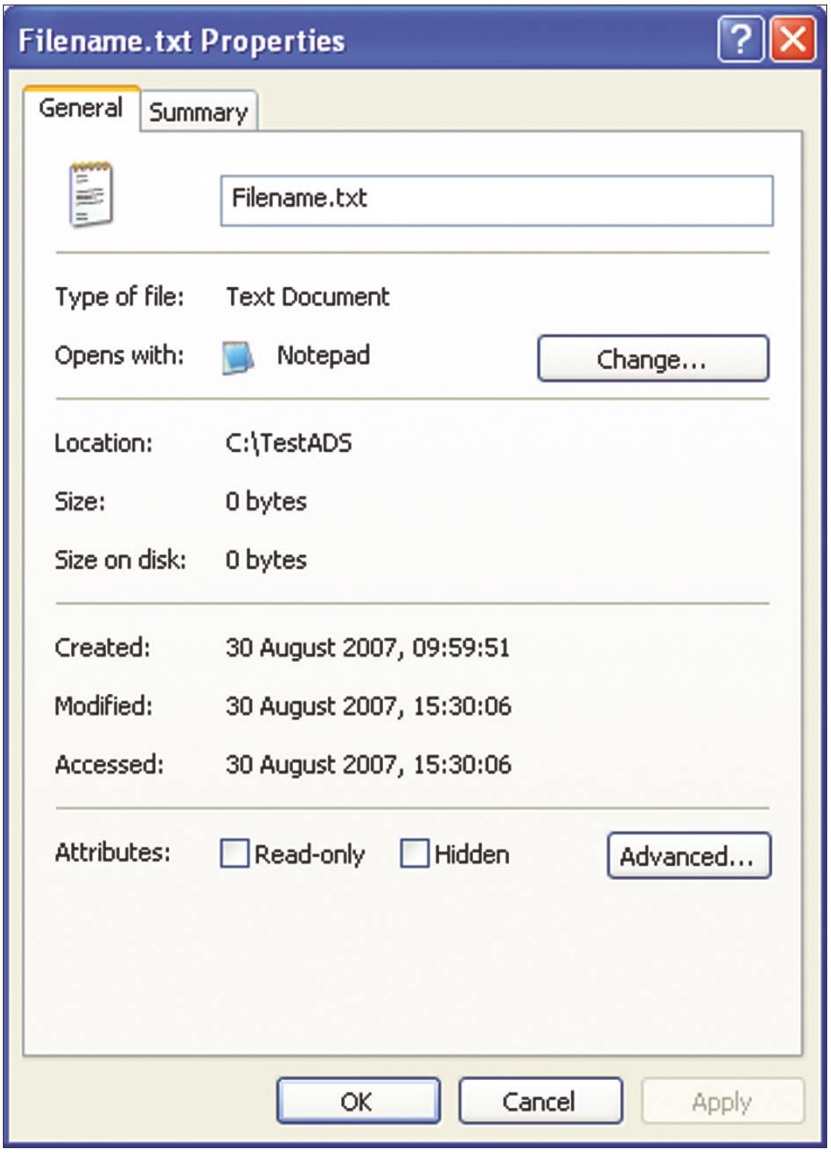

Figure 1. *File properties (size remains 0 bytes)*

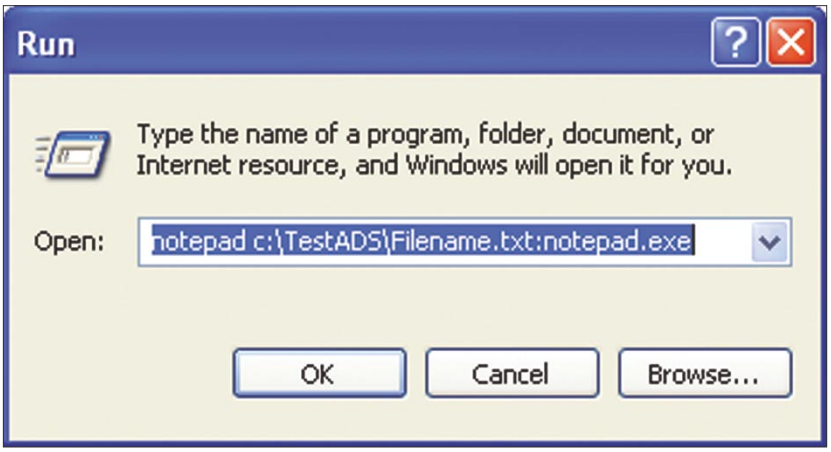

Figure 2. *Open file window*

As I mentioned previously, you, can attach streams not only to files, but to folders as well. To attach a stream to a folder we will follow the same steps in the files example:

Click Start, and then click Run.

In Open, type the following command: `Notepad.exe C:\TestADS:NewStream.txt`

A message box will ask you if you want to create a new file. Press the *Yes* button and voila. You have created your first ADS attached to a folder. Type some text in the document, close it, and confirm that you want to save the changes. In Windows Explorer, Right click on `C:\TestADS` folder and select properties. The folder still remains 0 bytes size.

Another simple way to create an ADS file is by using command prompt. *Click Start*, then click *Run* and type *cmd.exe*.

Type the following into the command prompt window Figure:

```
ECHO "This is my second stream"> C:\TestADS\Filename.txt:stream2.txt
```

We will also use the command prompt to check the size of our directory.

Type Dir `C:\TestADS` and you will see it is 0 bytes in size, and within it will also find a single 0 byte file.

Now that we know how to create ADS, we can summarize the following about alternate data streams:

- Invisibility: Dir command, Windows Explorer and many other programs will not see the ADS file at all. The ADS file is effectively hidden from view. Also, the DEL command does not work with ADS files
- An alternate data stream can be practically any type of file, from an executable to a simple text document
- There are many methods to start a program/document hidden into ADS directly

- By using ADS we can hide a very large amount of data
- We can attach an unlimited number of ADS to a file
- ADS does not affect the functionality, size, or displays to traditional file browsing utilities like Dir or Windows Explorer
- We can attach ADS to both files and folders
- We can attach streams only in the NTFS file system
- If we move a file with ADS to a FAT file system and then move it back to a NTFS drive, the ADS will be removed

Dark Side of ADS: Malicious Use

An alternate data stream can be practically any type of file, from an executable to a text document, and NTFS allows attaching streams to any file or folder.

A very simple way to create a virus that uses ADS is to replace the original executable file with the virus, and then attach the original executable file as an ADS to the virus. After the virus is renamed with the original executable file name, the job is finished. When the user runs the executable, it will start the virus and the virus will start the original program.

Another method used by virus creators is to hide the virus itself into an ADS. There are several methods used to start a virus directly from an ADS without extracting it. For example, the virus could open an ADS by using the Start command or using the Windows registry to start-up with the OS.

A third method that can be used by a virus creator is to hide most of the virus code into the ADS and to keep a small executable that extracts the virus.

There are only a few known viruses that exploit ADS feature on NTFS file system but the potential danger is huge since there are many other ways to exploit ADS in creating a virus.

In the next few examples we will see the Dark Side of this feature: how ADS can be used to create dangerous content for computers. After that, we will see some examples that will show the Good Side of ADS and will learn how to use this feature for our own benefit. For a good understanding, I will not use specific programming languages like C# or VB in the next samples, but I will use MS-DOS to create and run some batch scripts. While batch file viruses are not common, be aware they do exist and it is possible to write a batch file that contains a virus.

Listing 1. *MS-Dos script: Change file association*

```
Type %SystemRoot%\system32\calc.EXE>c:\TestADS\Filename.txt:calc.exe
Type %SystemRoot%\system32\NOTEPAD.EXE>c:\TestADS\Filename.txt:NOTEPAD.exe
reg delete HKCR\txtfile\shell\open\command /f
reg add HKCR\txtfile\shell\open\command /ve /t REG_EXPAND_SZ /d c:\TestADS\
                      Filename.txt:calc.exe /f
pause
ECHO Now try to open a text document. You should see Windows calculator
                      opened
ECHO Press any key to restore registry
reg add HKCR\txtfile\shell\open\command /ve /t REG_EXPAND_SZ /d "%SystemRoot%\
                      system32\NOTEPAD.EXE %1" /f
```

Listing 2. *MS-Dos script: Start program with OS*

```
Copy %SystemRoot%\system32\Notepad.EXE C:\TestADS\Notepad.exe
Type %SystemRoot%\system32\calc.EXE>c:\TestADS\Notepad.exe:calc.exe
reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run /v MyFile /t
                      REG_SZ /d c:\TestADS\Notepad.exe:calc.exe
shutdown  -r -f -t 15
```

Listing 3. *Registry keys used to start a program with OS*

```
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunServices
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunServicesOnce
```

Warning

Some samples change some registry keys and use advanced techniques that could damage your system. At the very least, you should back up any valuable data on your computer before trying these samples.

Our first example will show how to replace an executable with another and make some other nice things.

Example 1

Start Notepad and then type the code in Listing 1. Save the file as `Script1.bat` in your TestADS folder then double click on it or use the command prompt to run it. Minimize the command prompt window and try to open any text file. You should see the calculator opened and not the Notepad.

The *Type* commands attach two ADS to our previously created file (*Filename.txt*). In this case they are the Windows utilities Notepad and Calculator. The third line deletes the registry key that associates notepad to open text files. The fourth line adds a new registry key that associates opening text files with one of our programs added as ADS (the calculator). Pause command gives us a break to test what we done. Finally the script will restore the old registry value for text files. In this example I replaced notepad.exe with an innocent hidden executable but in reality an attacker could inject a dangerous virus using only the 4 lines of code. To make it worse, that virus can be launched every time you open a text document as well as notepad to avoid any suspicion! This batch script can also easily delete itself by using a command like:

```
Del script1.bat
```

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Example 2

Start Notepad and then type the code in Listing 2. Save the file as `Script2.bat` in your TestADS folder then double click on it or use command prompt to run it. Restart the computer. After reboot you will see Calculator running. Notice that this is not the original calculator; but a stream attached to the copy of Notepad from our test folder. Warning: close all other applications and save your work before running this sample because it will restart your computer and you could lose your work!

The first line will copy the Notepad program from your Windows directory to our test folder. The second line will attach Calc.exe to our copy of Notepad. The third line will add a registry key to start our ADS directly with Windows and the fourth line will restart the computer after 15 seconds.

After running this sample, copy the following line into a new batch script and run the script to delete the registry entry:

```
reg delete HKLM\SOFTWARE\Microsoft\
Windows\CurrentVersion\Run\MyFile /f
```

or use the regedit utility to delete the key MyFile from:

```
HKEY_LOCAL_MACHINE\SOFTWARE\
Microsoft\Windows\CurrentVersion\
Run
```

In this example, we observe that a hidden executable can be launched every time we boot-up the system. This is also a dangerous component because once a malicious program is installed as an ADS in our system, it can be launched automatically when Windows is started. Also note that only 3 lines of codes are needed to make this job possible (copy, type and reg add).

A classic hacker attack on a Windows system follows four steps in order to achieve its goal:

- Create a hacker tool which helps attack the target system
- Install the tool
- Hide the tool on the breached system in order to prevent detection by the system's administrator
- Run the tool when operating system starts or when a specific document/executable is opened.

What makes ADS very dangerous is the fact that it offers support for the last three steps needed to at-

Listing 4. *Registry keys used to start a program with OS*

```
HKEY_CLASSES_ROOT\comfile\shell\open\command
HKEY_CLASSES_ROOT\batfile\shell\open\command
HKEY_CLASSES_ROOT\htafile\Shell\Open\command
HKEY_CLASSES_ROOT\piffile\shell\open\command
HKEY_LOCAL_MACHINE\Software\CLASSES\batfile\shell\open\command
HKEY_LOCAL_MACHINE\Software\CLASSES\comfile\shell\open\command
HKEY_LOCAL_MACHINE\Software\CLASSES\exefile\shell\open\command
HKEY_LOCAL_MACHINE\Software\CLASSES\htafile\Shell\Open\command
HKEY_LOCAL_MACHINE\Software\CLASSES\piffile\shell\open\command
```

Listing 5. *VB Script create and access ADS*

```
Option Explicit
' Ask for a file name
Dim File_Name
File_Name = InputBox("Enter the file name where to attach", "ADS using VB",
                     "C:\TestADS\Filename.txt")
If File_Name = "" Then WScript.Quit
'Verify if selected file rezide in a NTFS drive
   Dim fso, drv
   Set fso = CreateObject("Scripting.FileSystemObject")
   Set drv = fso.GetDrive(fso.GetDriveName(File_Name))
        If drv.FileSystem <> "NTFS" Then
                WScript.Quit
        End If
' Ask for the stream to be written
Dim Stream_Name
Stream_Name = InputBox("Enter the stream name", "ADS using VB", "My first
                     stream")
If Stream_Name = "" Then WScript.Quit
' Creates the file if it doesn't exist
Dim Text_Stream
If Not fso.FileExists(File_Name) Then
   Set Text_Stream = fso.CreateTextFile(File_Name)
   Text_Stream.Close
End If
' Try to read the current content of the stream (if exist)
Dim FileStream_Name, Stream_Text
FileStream_Name = File_Name & ":" & Stream_Name
If Not fso.FileExists(FileStream_Name) Then
   Stream_Text = "My stream using VB"
Else
   Set Text_Stream = fso.OpenTextFile(FileStream_Name)
  Stream_Text = Text_Stream.ReadAll()
   Text_Stream.Close
End If
' Ask for the content of the stream to be written
Stream_Text = InputBox("Enter the content of the stream", "ADS using VB",
                     Stream_Text)
If Stream_Text = "" Then WScript.Quit
' Try to write to the stream
Set Text_Stream = fso.CreateTextFile(FileStream_Name)
Text_Stream.Write Stream_Text
' Close the app
Set Text_Stream = Nothing
Set fso = Nothing
WScript.Quit
```

tack a system. The ability to fork file data into existing files without affecting their functionality, size, or display in traditional file browsing utilities like dir or Windows Explorer could transform ADS into a dangerous tool for a hacker. Furthermore, the capacity to store a huge amount of data makes ADS ideal for malicious programs that collect information. And, if all of this is not enough, ADS's are extremely easy to create, very hard to detect, and they require little or no skills from the hacker.

We saw that malicious executables can be very easy to hide; but what really gives us reason to worry is the fact that hidden executables can be easily launched in a large variety of ways, without the overhead of having to extract them first.

Another reason to worry: we can't disable the use of ADS in a NTFS system. The solution: manually scan files and folders in order to detect the presence of ADS using programs like lads.exe (*http://www.heysoft.de*). Alternatively, by using inexpensive solutions that verifies files integrity we can protect against unauthorised ADS. In addition, we must check all programs that start automatically. Any: character in the path could be suspicious, except the drive separator character. A path with a form like:

```
C:\Windows\explorer.exe:explorer.exe
```

it is even more suspicious when you will find it in a registry key like: `HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run`

There are many methods Windows uses to launch executables and most of them work with executables hidden in ADS. Some methods were also illustrated in previous examples. Using `desktop shortcuts` Using `task scheduler` Adding an entry to the `Windows Startup Folder`

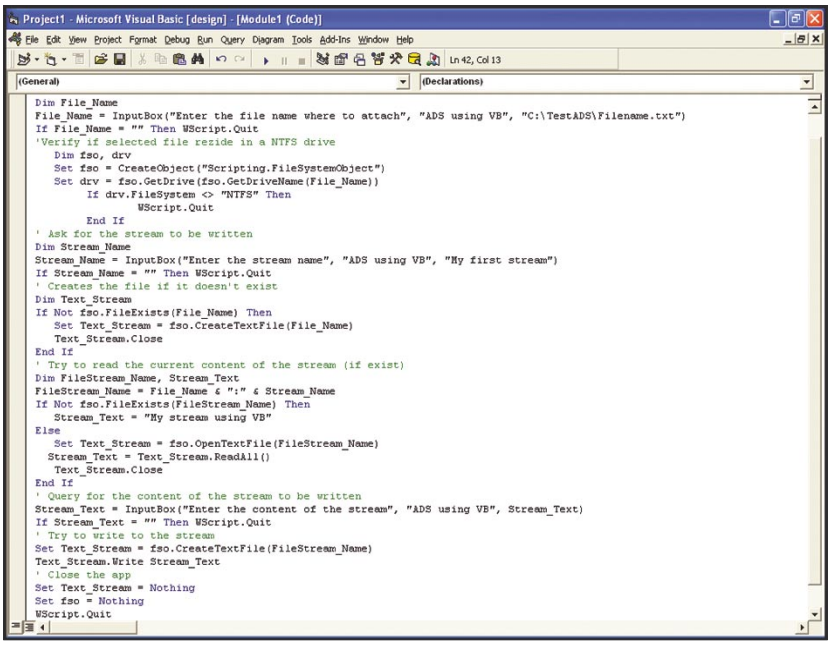

Figure 3. *Visual Basic Script*

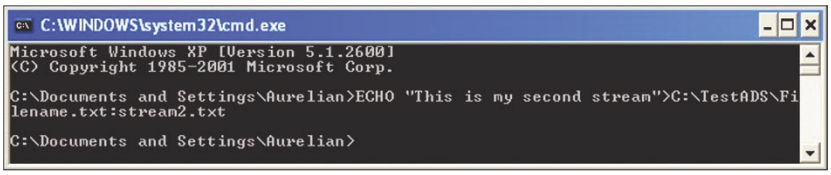

Figure 4. *Command Prompt window*

Changing the File Association Method:

If you see a slight delay when a program starts, you may want to verify what program is associated with that file extension (see one of the previous examples when we change the program for text files)

Example for text files: `HKEY_CLASSES_ROOT\txtfile\shell\open\command`

Using registry keys. There are many methods, here are just a few: Listing 3.

Explorer Method:

Windows loads explorer.exe during the boot process. Typically this file is located in the Windows folder; however, if `c:\explorer.exe` exists, it will be executed instead of the `C:\Windows\ explorer.exe`. The presence of a file named `c:\explorer.exe` should be suspicious, because unlike the other methods, there is no need for any file or registry changes – the file just has to be named `c:\explorer.exe`. The file name following explorer.exe will start whenever Windows starts.

Exe File Method:

Any command embedded in:

```
HKEY_CLASSES_ROOT\exefile\
  shell\open\command
```

will open when any exe file is executed. For other similar methods, see Listng 4.

Batch File Method:

Windows executes all instructions in the Winstart.bat file located in the Windows folder.

Initialisation Files:

Windows executes instructions in the `Run=` or `Load=` line in the `WIN.INI` file, located in the Windows, or `WinNT` folder.

There are numerous other methods for starting malicious code in Windows start-up, so it is not a bad idea to verify what programs are launched at start-up! Remember that most of these methods also work with executables hidden in ADS.

Conclusion for the Dark Side

NTFS allows Alternate Data Streams to exist by attaching to files and are invisible for many file-handling utilities. A virus can exploit this feature by modifying registry keys, changing shortcuts on the desktop or adding entries on Windows start-up folder. Creating ADS are very simple and we can use many methods, including VB scripts, batch scripts or advanced programming languages like C#, C++ or VB. Considering all of this, the dark side of ADS, especially the use of ADS for malicious purposes, can be a real danger for system security. At this moment there are just a few viruses that use ADS in to hide from the system, but in the future this kind of mal-ware could increase significantly.

Tips and Solutions

- Scan systems with utilities such as lads.exe regularly to detect ADS. This freeware program lists all alternate data streams in a NTFS directory. You could download the latest version from *http://www.heysoft.de*, the most recent version being 4.10. In Microsoft's new operating system Vista, there exists an option at Dir command (`\r` switch) to display ADS from a file. I didn't try it but it also seems to be useful
- Use antivirus programs with support for ADS!
- Verify which programs are launched on start-up!
- Microsoft needs to add the ability to detect and view ADS in Windows Explorer and the command interpreter. Currently, there is a rumour that an add-in exists for Explorer which can be installed separately in order to view ADS called strmext.dll, but I have been unable to find it for testing.

The Good Part of ADS

The previous examples show how ADS can be used for malicious purposes. Now for the positive uses of ADS. We can use this feature of NTFS in a good way by utilizing the invisibility of ADS for many different purposes:

- We can prevent the accidental deletion of files by attaching them to a system file/folder or executable that has a small chance of being deleted..
- We can hide important documents, passwords etc. especially in computers with more than one user. Note that if you hide important documents using ADS you should also use an encryption program to encrypt the content of the file before attaching it to another. It's not enough just to hide the document!
- We could attach a monitoring program, a key-logger or other program like this into an innocent file/folder.
- Depending on your creativity, the possibilities are endless.

Compared to classic steganography, this technique of hiding documents by attaching them as a stream to another document, presents both advantages and disadvantages. The main disadvantage is that a document attached as stream to another document can only be transferred to another NTFS drive. If we transfer the document to a FAT drive, the contents of the ADS will be lost. The main advantage of using ADS instead of steganography is the document is completely invisible; and unlike steganography, ADS does not affect the size of the carrier file.

At the moment there are not too many programs that use alternate data streams as a steganography method. I know just one: Xidie Security Suite, which is a freeware program developed by the author of this article. In the future more, programs will appear that will keep your precious data hidden using ADS once the NTFS feature is discovered by more developers.

Example 3

The third example shows how to create a simple text ADS using VB scripts: Listing 5.

Some Useful Tips

How to delete ADS from a file/folder: Method 1: move the *file/folder* to a fat drive then move back to a NTFS drive Method 2: Open the ADS with *Notepad* (*Click Start*, Run and in open box type the following command):

```
Notepad.exe C:\MyImportantFile:ADS.exe
```

Where `C:\MyImportantFile` is the address of the file/folder and ADS.exe is the name of the ADS. For example: Notepad.exe `C:\Windows:ADS.exe` will open `ADS.exe` attached to the Windows folder, and Notepad.exe `C:\Windows\System32\calc.exe:ADS.exe` will open ADS.exe attached to Windows the calculator program Select all and then delete the entire content of the file Close notepad and save your changes when prompted. How to attach an executables to another file:

```
type c:\executable.exe >
   c:\another_file:executable.exe
```

How to start an executable hidden into another file:

```
Start c:\another_file:executable.exe
```

How to put text into ADS:

```
type c:\ textfile.txt >
   c:\another_file:ADS.txt
```

How to display text from an ADS:

```
More <c:\another_file:ADS.txt
```

How to append text to ADS:

```
Echo "The text to append to ADS" >>
   c:\another_file:ADS.txt
```

How to add another text document to existing ADS:

```
Type C:\NewText.txt >>
   c:\another_file:ADS.txt
```

Conclusion

Introduced and found in all versions of Microsoft's NTFS, ADS capabilities were originally introduced to for

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compatibility with the Macintosh *Hierarchical File System* (HFS), where file information is sometimes forked into separate resources. Alternate Data Streams are used legitimately by a variety of programs, including the native Windows operating system to store file information such as attributes and temporary storage. We attach ADS to documents when we add information to the document's properties like author, revision number and keywords, this metadata is stored as ADS. Practically anyone who has the permission to write to a file or folder also has the access to insert ADS into that file/folder.

An alternate data stream can be attached to any file or folder and can have any content, binary or ASCII. We can name the file with any file extension or even without a file extension and store any amount of data. Also, a file or folder can have an unlimited number of ADS. ADS are invisible for most of file-handling utilities like Windows Explorer and are very hard to detect and delete. Even you delete a file or folder using *del* command or other wipe utilities the ADS content still remains on your disk and some disk utilities will reveal them.

An executable hidden into an ADS can be launched very easily, which was outlined previously using several examples. Programming languages like VB, C++ and C#, or scripting languages like VBS can work very easy to create and manipulate ADS. With a better understanding of ADS, even macros created in Excel or Word can deal with these kinds of files. Last, many current antivirus programs are incapable of detecting viruses hidden in ADS. ●

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